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Instructions

1. Suppose Y is a binary response variable, taking values either 0 or 1, while X_1 and X_2 are predictor variables. Predicting the probability1 / 1 point

$$\Pr(Y = 1|X_1, X_2)$$

involves setting up a

- ☐ Multiple linear regression model
- ☐ Simple linear regression model
- ☐ Poisson regression model
- ☒ Logistic regression model
- ☐ Gamma regression model

Correct

2. Which of the following statement(s) is/are true?1 / 1 point

(I) We say that a random variable Y follows the Poisson distribution if Y takes values in $\{0, 1, 2, \dots\}$ and

$$\Pr(Y = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

where $k = 0, 1, 2, 3, \dots$, for some $\lambda > 0$.

(II) If Y is a Poisson random variable, then the mean of Y is the same as the variance of Y .

(III) In Poisson regression, we assume that the mean of a Poisson random variable Y can be written as

$$E(Y|X_1, X_2, \dots, X_n) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n.$$

- ☐ I only
- ☐ II only
- ☐ III only
- ☒ I and II only
- ☐ I, II, and III

Correct

3. Which of the following component(s) is/are involved in a generalized linear model?1 / 1 point

(I) An exponential family model for the response variable y .

(II) A link function that connects the means of the response to the linear predictors.

(III) Predictor variables having a normal distribution.

- ☐ I only
- ☐ II only
- ☐ III only
- ☒ I and II only
- ☐ I, II, and III

Correct

4. Suppose we have the following Poisson regression model1 / 1 point

$$\ln(E(Y|X = x)) = \ln(\lambda(x)) = 0.23 + 0.05x$$

and consider x increases its value from x_0 to $x_0 + 1$. Compute the value of

$$\frac{\lambda_1 - \lambda_0}{\lambda_0} \times 100\%$$

where $\lambda(x_0) = \lambda_0$ and $\lambda(x_0 + 1) = \lambda_1$.

- ☐ 0.23%
- ☐ 23%
- ☐ 0.05%
- ☒ 5.127%
- ☐ 25.9%

Correct

5. Which of the following is the link function you should use when using the `glm()` function in *R*?1 / 1 point

(I) binomial, link = "logit"

(II) gaussian, link = "identity"

(III) Poisson, link = "log"

- ☐ I only
- ☐ II only
- ☐ III only
- ☐ I and II only
- ☒ I, II, and III

Correct